DOD-D-23443B
15 May 1978
Superseding
(see 6.5)

MILITARY SPECIFICATION

DI-(2 - ETHYLHEXYL) ADIPATE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of di-(2-ethylhexyl) adipate herein after referred to as the plasticizer.

2. APPLICABLE DOCUMENTS

2.1 Issues of document. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation - Packaging, methods of

STANDARDS

FEDERAL

FED-STD-313	Material Safety Data Sheets, Preparation and the Submission of

MILITARY

MIL-STD-129	Marking for Shipping and Storage
MIL-STD-1218	ACS Chemicals

Beneficial comments (recommendations, additions, deletions), and and pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 92 - 72	Flash and Fire Points by Level and Open Cup
ASTM D 94 - 71	Saponification Number of Petroleum Products
ASTM D 270 - 65	Sampling Petroleum and Petroleum Products
ASTM D 445 - 74	Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity)
ASTM D 1744 - 64	Water in Liquid Petroleum Products by Karl Fischer Reagent

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Traffic Department, 1616 P Street, Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIRMENTS

3.1 Clarity and color. The plasticizer shall be a clear liquid containing no visible suspended solid matter. The color shall be no darker than the color standard prepared in accordance with 4.5.1.

3.2 Physical and chemical properties. The physical and chemical properties of the plasticizer shall be in accordance with TABLE I.

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TABLE I. Physical and chemical properties.

PROPERTY	REQUIREMENT	
	MIN	MAX
Specific gravity, 25/25°C	0.921	0.929
Moisture, wt%	--	0.05
Acidity, wt%	--	0.01
Saponification number	300	304
Flash point, °C	196	--
Fire point, °C	216	--
Kinematic viscosity, 38°C, mm^2/s	8.16	8.76

3.3 Material safety data sheets. The contractor shall prepare and submit material safety data sheets in accordance with FED-STD-313 as specified in the contract (see 6.4).

3.4 Workmanship. The material shall be uniform, free from contamination, foreign material or any other defect that would prevent its use for the purpose intended. The material shall be manufactured in accordance with standard manufacturing procedures of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following laboratory conditions:

a. Temperature: Room ambient 18 to 35°C (65 to 95°F)
b. Altitude: Normal ground
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum.

4.3 Sampling

4.3.1 Lot. Unless otherwise specified in the contract (see 6.2), a lot shall consist of all plasticizer manufactured in one continuous production run or in one batch, under essentially identical conditions, from the same raw materials, and to be offered for acceptance at one time.

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Several batches manufactured from the same raw materials, may be blended to form a uniform larger batch which shall then constitute a lot for inspection purposes.

4.3.2 Sampling for quality conformance inspection. Samples for quality conformance inspection shall be obtained in accordance with the following:

4.3.2.1 Containers. The number of containers to be sampled shall be in accordance with TABLE 3 of ASTM D 270-65. Primary samples shall be obtained from the sample containers in accordance with the tube sampling method of ASTM D 270-65. Each primary sample shall consist of approximately 0.5 litres (L) of plasticizer. A composite sample of approximately 1.0 L shall be prepared from equal portions of each primary sample, placed in airtight glass bottles, and labeled with the material name, manufacturer, plant, contract and lot numbers, and lot size. All primary sample remnants shall be held for possible future reference.

4.3.2.2 Tank cars. Each tank car shall be sampled in accordance with the bottle or beaker method of ASTM D 270-65. The sample shall consist of approximately 1.0 L of plasticizer and shall be placed in an airtight glass bottle labeled as specified in 4.3.2.1.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the tests and examinations detailed in 4.5 performed on samples selected in accordance with 4.3.2. Failure of the composite sample to pass any test or examination shall result in rejection of the lot.

4.5 Tests and examinations. Unless otherwise specified herein, all chemicals shall be ACS grade in accordance with MIL-STD-1218.

4.5.1 Color. The color shall be tested as follows:

a. Transfer 1.00 ± 0.01 gram(g) of iodine and 5.0 ± 0.1 g of potassium iodide to a 150-mL beaker. Add 25 mL of distilled water to dissolve the iodine and potassium iodide. Transfer the solution quantitatively to a 1000-mL volumetric flask, dilute to the mark with distilled water, and mix thoroughly.

NOTE: Do not use solution after 24 hours.

b. Fill a clean 50-mL Nessler tube to the mark with sample plasticizer. Fill another 50-mL Nessler tube to the mark with the standard color solution (a. above). Hold the tubes side by side and compare their colors using transmitted light while looking downward through the tubes.

c. The sample shall be rejected if the color is darker than that of the standard solution.

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4.5.2 Clarity and workmanship. The clarity and workmanship shall be examined by inverting a bottle of the sample several times and visually examining the plasticizer. The presence of any suspended solid material shall result in rejection of the sample.

4.5.3 Specific gravity. Determine the weight capacity to the nearest 0.1 mg of a 25-mL Hubbard pycnometer by filling with distilled water at 25 ± 1°C. Fill this calibrated pycnometer with a portion of the composite sample which has been cooled to below 25°C. Insert the pycnometer cover, taking care not to introduce air bubbles, and condition the pycnometer in a water bath at 25 ± 1°C for a period of not less than 30 minutes. Remove the droplet of sample from the overflow capillary of pycnometer, wipe the outside surface of the pycnometer dry, and weigh to the nearest 0.1-mg. Calculate the specific gravity as the ratio of the weight of sample used to the water capacity of the pycnometer.

4.5.4 Moisture. The moisture content shall be determined in accordance with ASTM D 1744-64 except that the titration solvent shall be anhydrous methanol, the sample size shall be 10 mL and the calculation shall be as follows:

$$\text{water, wt\%} = \frac{CF}{10W}$$

where:

C, F and W are as defined in ASTM D1744-64 and 10 is a factor for conversion from mg of water per g of sample to wt% water.

4.5.5 Acidity. To 50 mL of absolute ethanol in a 125 -mL flask, add 2 drops of 1-percent phenolphthalein indicator. Titrate with 0.1 normal sodium hydroxide until a faint pink color can be observed in the flask. This color shall remain pink for not less than 30 seconds. Determine the weight of the flask and its contents to the nearest 0.1 g. Add about 25 mL of sample plasticizer and reweigh to the nearest 0.1 g to determine how much of the material was added to the flask. Titrate with 0.1 normal sodium hydroxide solution until a faint pink color can be observed in the flask. This color shall remain for at least 30 seconds. Calculate the percent acetic acid as follows:

$$\text{Percent acetic acid} = \frac{6.0\ V\ N}{W}$$

where:
V = Volume of sodium hydroxide to titrate sample solution, mL
N = Normality of sodium hydroxide
W = Weight of sample, g

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4.5.6 Saponification number. The saponification number shall be determined in accordance with ASTM D94-71 except that a 2-gram sample and a 3-hour reflux time shall be used.

4.5.7 Flash point and fire point. The flash point and the fire point shall be determined in accordance with ASTM D92-72.

4.5.8 Kinematic viscosity. The kinematic viscosity in mm^2/s (see 6.3) at 38°C shall be determined in accordance with ASTM D445-74 using a Ubbelohde viscometer.

4.5.9 Packaging inspection. The packaging, packing and marking shall be inspected to verify conformance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging and packing. Unless otherwise specified in the contract (see 6.2), packaging and packing shall be level C.

5.1.1 Level C. Unless otherwise specified in the contract (see 6.2), packaging and packing of the plasticizer shall be in accordance with Method III of MIL-P-116. The packaging and packing shall be of such construction and materials that the contents will be adequately protected against loss or contamination. Container size shall be as specified in the contract (see 6.2). Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), each container shall be marked in accordance with MIL-STD-129. Marking shall include but not be limited to, the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and location.

c. Material trade name.

d. Net weight and volume.

e. Lot number, batch number (s), and date of manufacture.

f. Storage conditions.

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g. Toxicity precautions, if applicable.

h. Contract number.

6. NOTES

6.1 Intended use. Di (2-ethylhexyl) adipate in accordance with this specification is intended to be used as a plasticizer in propellant and explosive compositions.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Quantity required.
c. Inspection conditions if other than as specified (see 4.2).
d. Lot size if other than as specified (see 4.3.1).
e. Packaging requirements if other than as specified (see 5.1).
f. Size of container required (see 5.1.1).
g. Any special marking required (see 5.2).
h. Material safety data sheet requirements (see 3.3 and 6.4).

6.2.2 Contract data requirements. Any data required for delivery in connection with this document shall be specified on a DD Form 1423 incorporated into the contract. Such data will be delivered as identified on completed (numbered) DIDs (Data Item Descriptions/DD 1664) which will be documented in the Applicable ADL (Authorized Data List).

6.3 Definition. The International System of Units (SI) recognizes the m^2/s as the official metric unit for kinematic viscosity. One centistoke (cSt) is therefore equal to one mm^2/s.

6.4 Material safety data. Material safety data sheet requirements are applicable and should be specified in the contract as required by the Armed Services Procurement Regulations (ASPR) 1-323.2.

6.5 Supersession data. This specification supersedes MIL-P-23443A (WEP), dated 22 April 1965, with Amendment 1, dated 7 July 1977; and MIL-D-13383(Ord), dated 19 April 1954. This specification also includes the requirements of and supersedes Naval Sea Systems Command purchase description WS 2237, dated 11 October 1962.

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6.6 Changes from previous issue. Due to the comprehensive nature of this revision, the margins of this specification have not been marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. Bidders and contractors are advised to evaluate the requirements of this document based on the entire content irrespective of the relationship to the last previous issue.

Custodians:
Navy - OS
Army - AR
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Review Activities:
Army - MI
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Preparing Activity:
Navy - OS

Project No. 6810-B108

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

DOD-D-23443B, DI-(2 - ETHYLHEXYL) ADIPATE, TECHNICAL

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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